

ID S1**Temperature sensor with 1-wire interface (DS18B20)****Prior knowledge**

- How to prepare, carry out, and document laboratory experiments
- Basic terms and definitions of sensor technology
- Temperature sensors
- Function of Peltier elements
- 1-wire interface
- Programming the Arduino (Nano or equivalent) under the Arduino IDE
- Programming in Python (Anaconda/Spyder)
- Data Sheet Temperature Sensor (<https://www.analog.com/media/en/technical-documentation/data-sheets/ds18b20.pdf>)
- Data Sheet Peltier Element (<https://cdn-reichelt.de/documents/datenblatt/C800/TEC12705-1.pdf>)
- Basic knowledge of Python and the libraries is an advantage (e.g. <https://py-tutorial-de.readthedocs.io/de/python-3.3/>, <https://www.python-kurs.eu/numpy.php>, ...)

Material

- Arduino Nano
- Temperature sensor with 1-wire interface based on DS18B20
- Breadboard
- USB Mini Cable
- 3 male-male Connectors
- Peltier element as heat source / heat sink
- Laptop, Arduino-IDE, Python-IDE (Anaconda), simple terminal software (z.B. HTerm)
- Arduino IDE: one wire library (reinstall if necessary)

OneWire by Jim Studt, Tom Pollard, Robin James, Glenn Trewi...

2.3.8 installed

Access 1-wire temperature sensors, memory and other chips.

[More info](#)

2.3.8

REMOVE

- Oscilloscope

Duration: 3h

Setup:

Hardware:

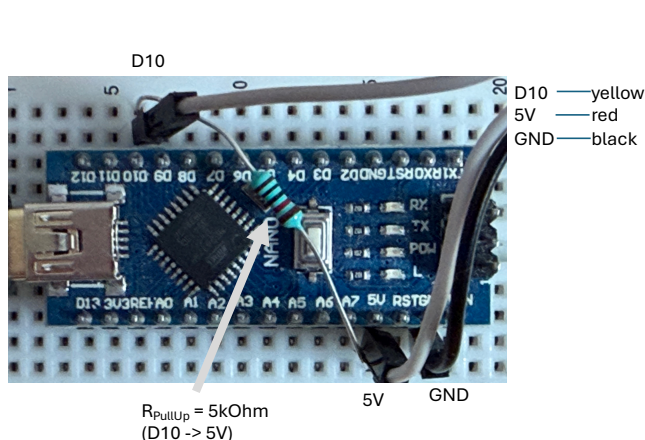


Figure 1: Electrical setup

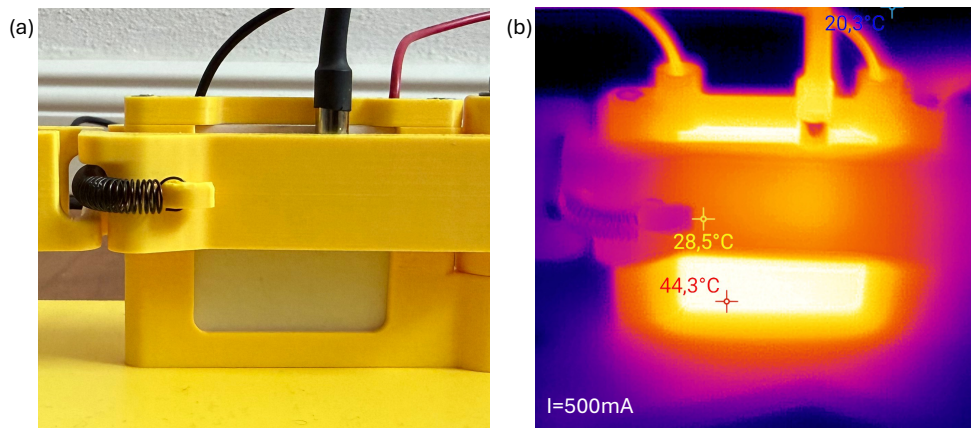


Figure 2: (a) Mechanical setup with Peltier element, (b) Thermal camera image

Software:

- Arduino IDE
- Anaconda/Spyder, ...
- Prepared programs from the website (Arduino, Python):
 - TemperaturSensor.ino (Arduino)
 - 01_Read_USB_CSV_TimeStamped.py (Python, recording of measurement data with time stamp, generates csv file)
 - 02_data_plotter.py (Python, reads csv file and creates figuresy)
 - 03_function_plotter_timeseries.py (Python, creates function plot for the fit-function)

Part 1 – Setup and basic operation

- Set up the Arduino and the temperature sensor on the breadboard
- Three connections are required: 5V, GND, DQ. Make sure, the pull-up resistor is correctly connected
- Download the program "TemperaturSensor.ino" from the Teams drive, compile it and transfer it to the Arduino
- Use "Tools/Serial Monitor" or "Tools/Serial Plotter" to check the sensor data; make sure that transmission parameters are set correctly (especially the baud rate) – What happens if you change the baud rate (Arduino / PC)?
- Download the program "01_Read_USB_CSV_TimeStamped.py" and download it into your editor under Anaconda/Spyder
- Start the program there and check the output values, you may have to install the libraries "pyserial", "numpy", "time" and "matplotlib" in Anaconda Navigator beforehand
- What changes when you change the temperature by touch? Document your result.

Deliverables:

- Summary of the task in your words
- Result, confirmation of the function (photo of the set-up, screen dump, sensor output when touching the temperature sensor)
- Create a block diagram of the structure (HW) including the electrical supply to the components and the interfaces
- Analyze both programs: How is the measurement carried out and how do the measurement results get to the laptop? How is the timestamp generated? Create a data flow diagram based on your analysis.

Part 2 – Investigation of the 1-wire interface

- Check the oscilloscope probe for correct calibration, calibrate if required
- Connect the probe to the DQ pin
- Measure voltage relative to GND as a function of time

- Record at least five screen dumps with different timescales between 500ms/div and 500us/div to see how often telegrams are transmitted and document the start of a communication in detail.
- Annotate at least the first 5 bytes of a telegram after the Start Of Frame (setting 500us/div). Which command is transmitted?

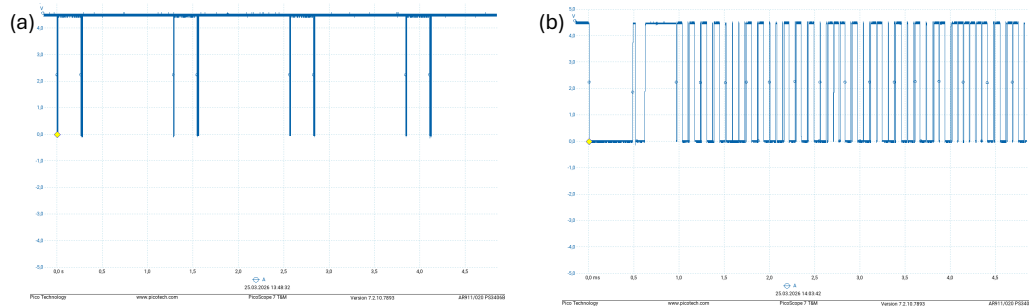


Figure 3: Examination of the interface (DQ) with the oscilloscope (a) 500ms/div (b) 500us/div

Deliverables:

- Summary of the task in your words
- Photo of the experimental setup
- At least five screen dumps with the above timescales,
- Annotated screen dump at high resolution. Explanation of the values transferred.
- How often is a new temperature value transferred to the bus?

Part 3 – Determination of the time behaviour $T(t)$

- Watch the video of a thermal camera on the Teams Drive showing how the Peltier element heats up (you can fast-forward!). Estimate the required measurement time so that you can determine the time constant.
- Start the measurement to record the time series at room temperature and initially keep the current through the Peltier element at 0A
- Abruptly change the current (step function, switching on the current source) to 0.5A and measure the timing of the temperature on the sensor
- Approximate the behavior by $T(t) = T_0 + \Delta T(1 - \exp(-(t - t_0)/\tau))$
- Determine T_0 , t_0 and τ and evaluate your result: How well does the formula match the measurement? What is the maximum observed deviation?

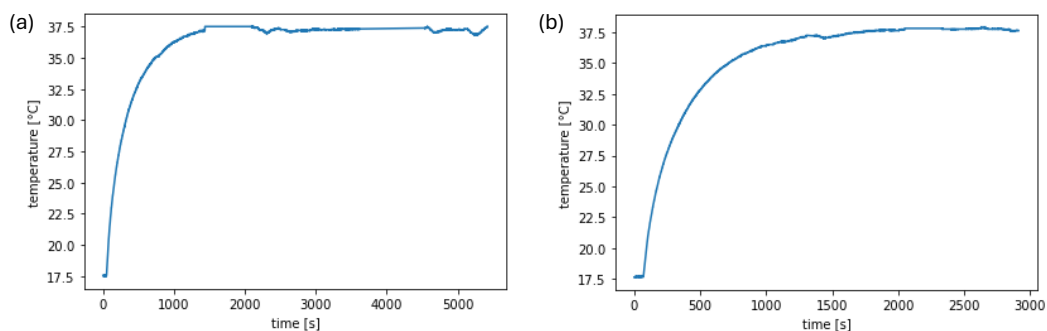


Figure 4: Exemplary recorded time series for $I=500\text{mA}$

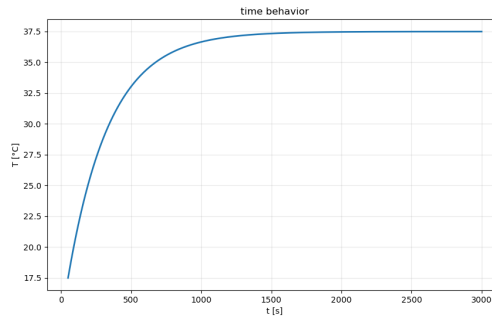


Figure 5: Fit of the specified function to the measurement data

Deliverables:

- Summary of the task
- Photo of the experimental setup
- Time series of your measurement
- Determination of T_0 , t_0 and τ
- Discussion of errors and determination of the maximum deviation between measurement data and approximation

Experimental part 4 – Determination of the characteristic curve $T(I_{\text{Peltier}})$ for this setup

- Allow the Peltier element to cool to room temperature
- Record a characteristic curve $T(I_{\text{Peltier}})$, starting at room temperature and $I_{\text{Peltier}}=0\text{A}$ ascending to $I_{\text{Peltier}}=1\text{A}$. Wait long enough for each measurement to allow thermal equilibrium to be established.

Deliverables:

- Task in your words
- Documentation of the respective waiting time and table $T(I_{\text{Peltier}})$
- Graph $T(I_{\text{Peltier}})$
- Discussion of possible sources of error for this measurement